Movie Recap Tool

How the app works: features, models, user flow and the rendering pipeline

Upload a short recap clip or a full movie/drama. The app transcribes the speech, translates it to Burmese, records a Burmese voice-over (Gemini TTS), draws Burmese subtitles, blurs the original subtitles and watermarks, applies the chosen look and picture format, and lets the user edit and re-render the result.

Part	What it is	Where
User app	Create video page + usage dashboard + editor (React 18, Vite, Tailwind, Myanmar / English)	frontend/ (port 3000)
API	FastAPI: accounts, quotas, upload, job status, signed file URLs, editor re-render	backend/main.py
Workers	One job each: detect, transcribe, voice, fit, render. RQ + Redis, or an in-process thread pool in dev	backend/pipeline.py, jobs_queue.py
Owner dashboard	Separate app on its own sub-domain: users, jobs, voice models, audit log, settings	admin/ (port 3001), backend/admin/
Storage / DB	Local disk or S3-compatible (Spaces); SQLite in dev, Postgres in production; files deleted after 24 h	storage.py, db.py

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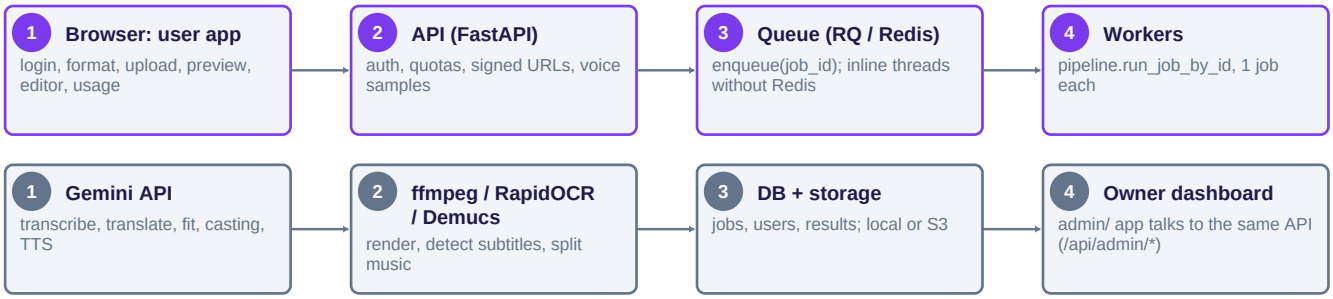
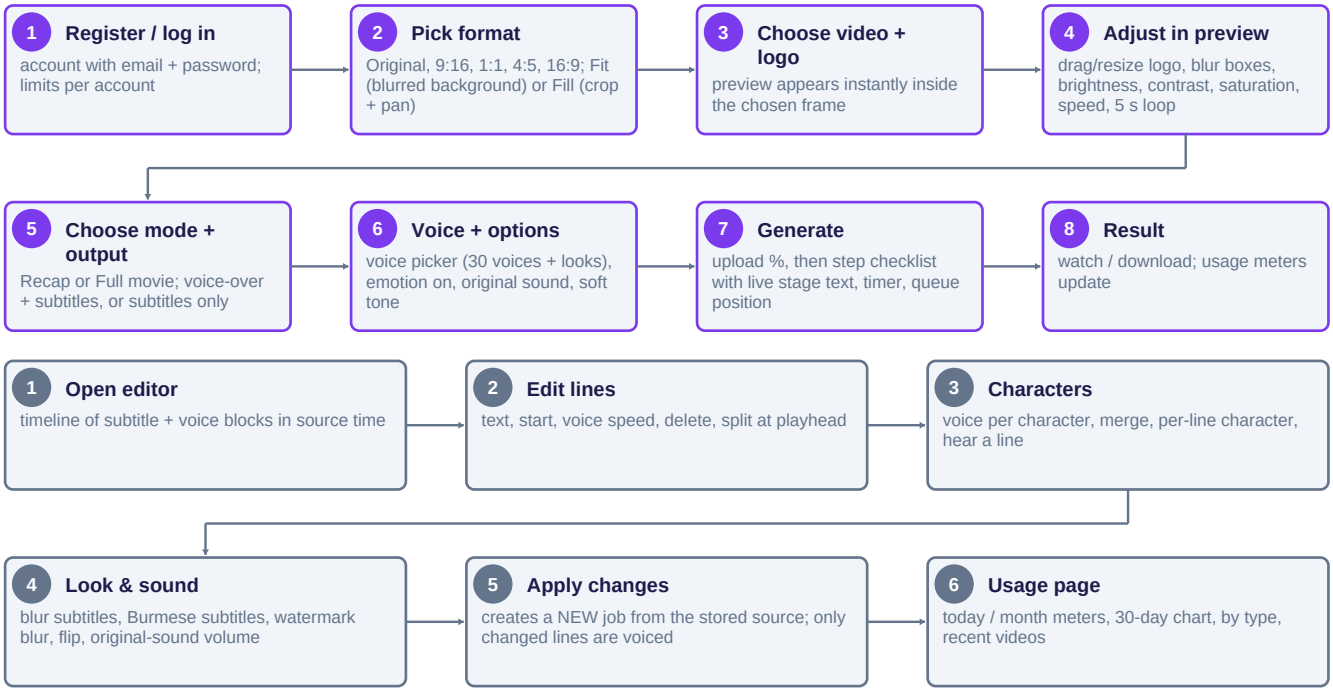
1. Architecture



Without **REDIS_URL** the API runs jobs itself in a thread pool (development). With it, separate worker processes pick jobs up; the API only enqueues. Result files go to storage; the API returns signed, expiring URLs.

2. User flow



Navigation: fixed sidebar on wide screens (Create video, My usage, today/month meters, account, language, log out); on phones a top bar with a menu button opens it as a drawer. The Create page stays loaded while the usage page is open, so a running job is not lost.

3. Features and functions

Upload form

- **Two modes.** Recap video (one narrator voice) and Full movie / drama (several characters, each with a fixed voice).
- **Output choice** in both modes: Burmese voice-over + subtitles, or Burmese subtitles only (original audio and timing untouched).
- **Format before upload** (like CapCut): Original, 9:16, 1:1, 4:5, 16:9. Fit = whole picture over a blurred copy; Fill = crop, drag to pan.
- **Live preview** without any request: brightness / contrast / saturation, speed, flip, draggable + resizable logo (corner handles, size slider), up to 6 blur boxes, seek bar, 5-second loop.
- **Voice picker**: 4 voices per gender plus a full library (30 Gemini voices + 32 character looks) with play buttons and search; free pre-generated samples.
- **Emotion** directions on key lines (angry, sad, scared...), on by default. **Original sound**: whole track lowered, or music/effects only (Demucs), or off. Optional generated soft tone.
- **Time estimate**: queue position and ETA; Cancel at any step (nothing is charged for cancelled or failed jobs).

Processing features

- Scene-cut detection; transcription and translation to colloquial Burmese in parallel chunks; per-character labels, ages and genders (Full mode).
- Source-subtitle blur only while the subtitle is on screen; watermark detection; manual blur boxes in output-frame coordinates.
- Script fitting: each voice line is shortened or lengthened to match the original speech length; voice speed-up is the last resort (max 1.35x), then video slow-down (min 0.85x), then freeze frame.
- Voice clip cache (voice + text + style) so edits only pay for changed lines; one-line voice preview in the editor.
- Failure tolerance: a refused line goes down a ladder (no emotion, other model, softer wording, silent clip + warning); a job fails only if the service is really down.

Accounts, limits and admin

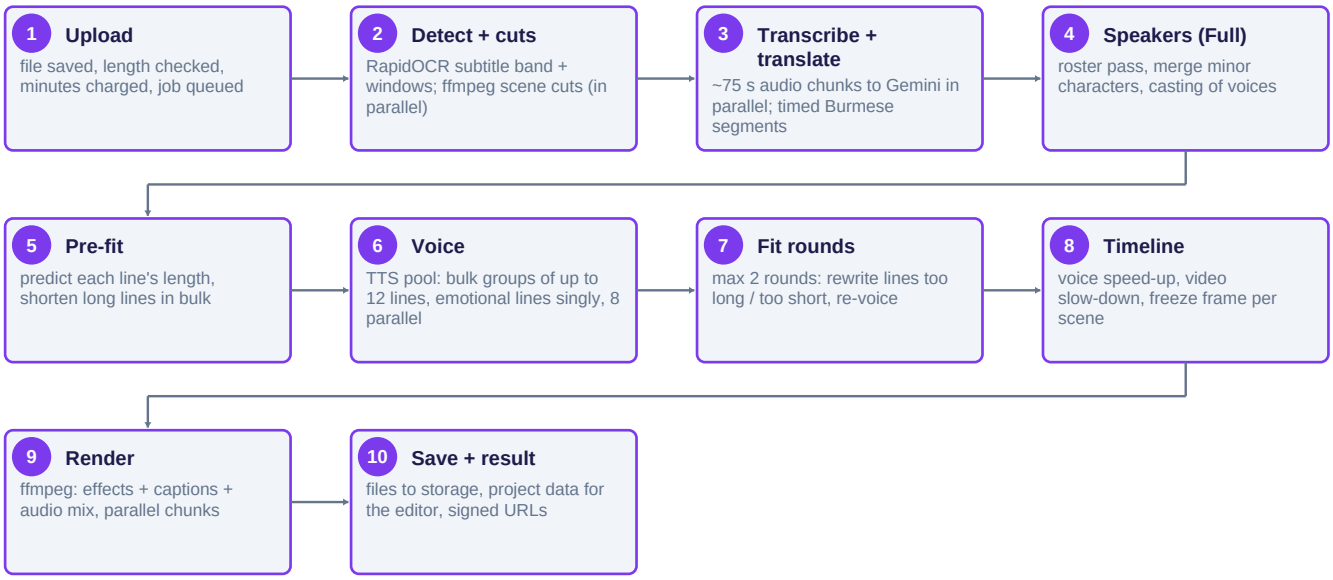
- Per user: **100 minutes/day** and **2000 minutes/month** (UTC), refunded when a job fails or is cancelled; re-renders are free. Upload is refused (HTTP 402) when either is used up.
- Owner dashboard: overview, users (create, edit limits, enable/disable, reset password), jobs (filter, cancel), voice-model usage and Google quota, audit log, maintenance mode and global daily ceiling.

4. Models and services used

What	Model / tool	Used for	Notes
Text + audio understanding	Gemini **gemini-3.8-flash** (GEMINI_DUB_MODEL)	Transcription + Burmese translation, speaker roster, casting, script shortening/lengthening, merge of characters	Thinking off (about 4x faster). Quality of the newest prompts is UNVERIFIED on real films
Voice (TTS), AI Studio pool	gemini-3.1-flash-tts-preview, gemini-2.5-flash-preview-tts, gemini-3.8-flash-tts, gemini-3.8-flash-lite-tts, gemini-2.5-pro-preview-tts	Burmese voice-over, one model per video	Tier 1: 10 requests/min and 100/day per model; the pool switches model when one is used up. Flash-Lite does not support Burmese
Voice (TTS), Google Cloud	gemini-2.5-flash-tts, gemini-2.5-pro-tts via Cloud Text-to-Speech	Being tested as a second provider (not wired into the app yet)	Same voices; separate quota; tested with 6 parallel requests OK; Burmese judged acceptable but not as good as AI Studio
Script fitting (optional)	Claude (FIT_TEXT_MODEL, default claude-haiku-5-5) if ANTHROPIC_API_KEY is set, else Gemini	Rewrite lines to fit their time slots	Batched ~20 lines per request
Subtitle / watermark detection	RapidOCR	Find the original subtitle band, line height, watermarks	<=16 sampled frames, then a band-only scan for time windows
Music / effects separation	Demucs (no_vocals stem)	Original sound without speech	Slow (about 0.43x real time on M2 Air); cached; may remove screams as 'vocals'
Rendering	ffmpeg (filter_complex)	Reframe, blur, grade, speed, logo, captions, audio mix	Long videos render in parallel chunks, then join
Fallbacks	ElevenLabs Dubbing, local faster-whisper + NLLB-200	Only when DUB_FALLBACK=1	Expensive / weak Burmese; last resort
Duration model	duration ~ a x syllables + b, fitted per job	Predict voice length without a request	corr 0.89 on real clips; also in the editor (JS)
Voice pitch check	numpy F0 (pitch.py)	Re-voice emotional lines that drift more than 4 semitones	

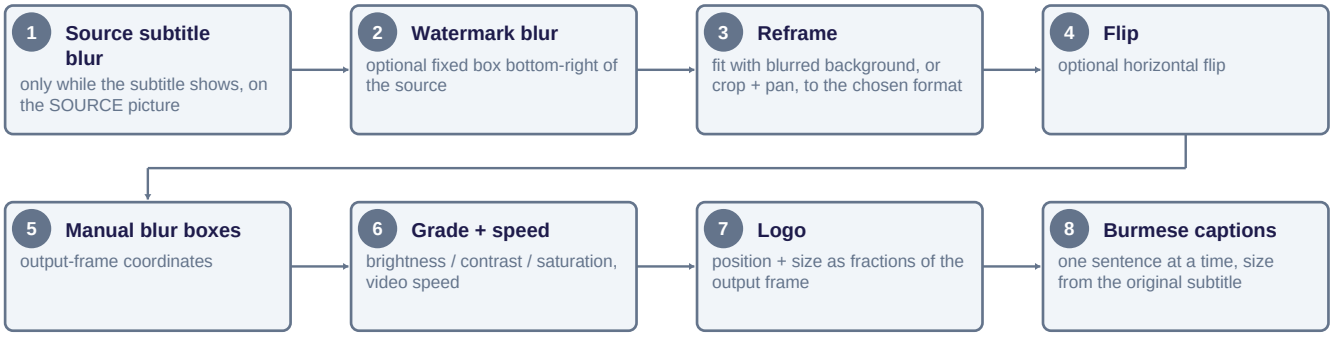
Python backend: FastAPI, SQLAlchemy, RQ, google-genai, scrypt + JWT for accounts. Frontend: React 18, Vite, Tailwind, axios.

5. Rendering pipeline for one job (pipeline.py -> recap_engine.py)



Progress shown to the user: 3% detect, 10% translate, 35-60% voice, 62-67% fit, 68% timeline, 75-100% render.

ffmpeg render order (recap_engine._render_single)

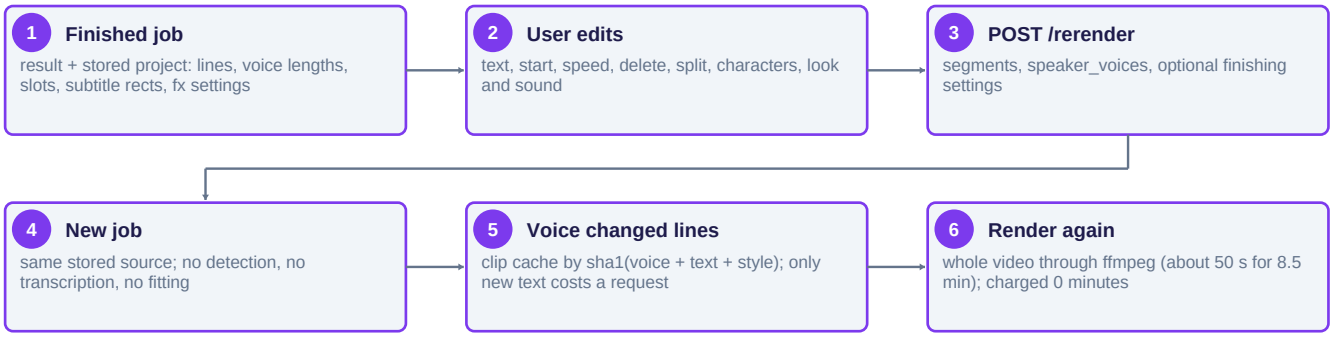


Audio: voice clips placed on the timeline; original sound (if kept) mixed under the voice at 0.8 (0.7 when the whole original track is kept), sidechain ducking ratio 4, user volume slider 0-1.5. Output size: short side = min(1080, source short side), never upscaled.

Time budget

Product goal: a 10-minute video in at most 10 minutes. Measured floor with fake Gemini on an M2 Air: 95 s (render 82 s, detection 8 s). Budget = clamp(0.9 x video length, 150 s, 540 s) with a render reserve; pre-fit and each fit round start only if budget is left. Hard limit 720 s: the job fails cleanly and is refunded. Real runs so far: 502 s video in 369 s, 508 s video in 380 s.

6. Editor and re-render flow



- The editor shows a live estimate for edited text (voice length vs slot, silence, total length) using the duration model in JavaScript, with no request.
- 'Hear voice' on a selected line makes one TTS request in the character's voice, and the clip is cached so Apply does not pay again (12 previews/min/user).
- Source uploads are kept for FILE_TTL_HOURS so re-render works; results are deleted after 24 h.
- Idea not built: cache rendered ~40 s windows and re-render only the changed ones; undo; merge lines; waveform/thumbnails.

7. Limits, quotas and cost

Topic	Value	Notes
Per user	100 min/day, 2000 min/month	UTC day / month; remaining = the smaller of the two; failed and cancelled jobs refunded; re-renders free
Gemini TTS Tier 1	10 requests/min and 100/day per model	5 models in the pool = about 450 voice requests/day; a 10-min video needs about 65 Gemini requests
Cost estimate (UNVERIFIED)	about 0.55-0.8 THB per video minute	2000 min/month = about 1,100-1,600 THB at the cap; TTS about 1.4 US cents per speech minute; not measured per job yet
Capacity estimate	30,000 video-min/day needs 10-20 busy cores (40-80 at peak)	A 2 vCPU droplet is for a small beta only
Global controls	maintenance mode, global daily ceiling, active-job limit per user	Changeable at runtime in the owner dashboard

8. API endpoints

Method + path	Purpose
POST /api/auth/register, /login; GET /api/auth/me	Accounts; me returns limits, usage, is_admin
GET /api/usage	Own usage: today/month, 30 days, by type, recent videos, reset times
GET /api/voices, /api/voice-samples/{file}, /api/looks, /api/look-samples/{file}	Voice picker data and free samples
POST /api/jobs	Upload + start a job (mode, output, voice, format, logo, blur boxes, look settings); 402 when over limit
GET /api/jobs/{id}	Status, progress, queue position + ETA, result, project data
GET /api/files/{name}	Signed, expiring result / preview files
POST /api/jobs/{id}/rerender	Editor Apply: new job from the stored source
POST /api/jobs/{id}/preview-line	Hear one line in a character's voice
POST /api/jobs/{id}/cancel	Stop a queued or running job
/api/admin/* (admin only)	overview, users, jobs, tts, google-quota, settings, audit

9. Open items and unverified parts

- **Not verified in a browser:** sidebar / drawer / usage dashboard layout, chart hover, preview touch dragging, the timeline on a phone, the owner dashboard UI.
- **Not verified with real data:** the rewritten translation prompt, full-mode characters/casting on a long film, looks, 1080-wide render time against the 95 s floor, Burmese UI wording (AI-written, needs a native review), Google Cloud quota matching.
- **Capacity:** AI Studio Tier 1 cannot serve many users. Google Cloud TTS (separate quota) is being tested as an added provider; it is not connected to the app yet.
- **Cost:** real tokens / voice seconds per job are not logged yet, so cost per minute is an estimate.
- **Known limits:** Demucs treats screams and ghost voices as vocals and removes them; watermarks that move are not handled; auto-detected blur boxes are not shown in the preview; manual blur boxes are not time-limited.
- **Not built:** payments / plans, login / failed-admin logging, deleting users, Docker deployment is untested, legal terms and takedown process for uploaded footage.